

$$2\sqrt{x-3} = w\sqrt{x}$$

$$4(x-3) = w^2x$$

$$4x - 12 = w^2x$$

$$12 = 4x - w^2x$$

$$12 = x(4 - w^2)$$

$$x = \frac{12}{4 - w^2}$$

(d) Two complex numbers are defined by $u = 1 + pi$ and $v = 5 + 3i$.

Given that $\arg\left(\frac{u}{v}\right) = \frac{\pi}{4}$, find the value of p .